

Zach Taylor

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EDUCATION

Oregon State University

Bachelor of Science in Computer Science

September 2019 – June 2023

Corvallis. OR

EXPERIENCE

Software Engineer Intern *Rohde & Schwarz*

April 2024 – Jan 2025

- Enabled advanced network analysis for clients like the U.S. Army, Tesla, and emergency services by delivering custom scanning features and frequency band configurations designed, implemented, and integrated using **C++**.
- Improved processing speed by **25%** and code maintainability by redesigning internal tools with smart pointers and more efficient lookup algorithms.
- Ensured stability and reliability across multiple software components by uncovering edge cases through rigorous unit testing and debugging with **Google Test**.
- Fostered team alignment and collaboration while ensuring software development lifecycle compliance by leading weekly development meetings and coordinating cross functional efforts.

Undergraduate Teaching Assistant *Oregon State University*

September 2020 – June 2023

- Strengthened students' foundational and practical skills by teaching **Data Structures** and **Operating Systems** in **C**, and guiding full stack web development using **JavaScript**, **Node.js**, and **HTML/CSS**.
- Raised academic performance by **30%** among struggling students by delivering personalized one-on-one and small group tutoring focused on reinforcing core concepts and addressing individual challenges.
- Improved students' problem solving abilities and debugging proficiency by walking through real code issues and encouraging a deeper understanding of troubleshooting techniques.

Software Development Intern *Portland State University*

June 2019 – September 2019

- Achieved **97%** accuracy in object detection by implementing a model with **Python OpenCV** for image capture and leveraging TensorFlow's transfer learning APIs.
- Delivered a versatile image processing solution by configuring the model to handle both live camera feeds and static image datasets for real time and batch applications.
- Led collective goal setting and streamlined project execution by managing weekly deliverables and fostering innovative problem solving through open collaboration.

PROJECTS

Patient Monitoring Robot | *Python, OpenCV, ROS2*

[GitHub](#)

- Enhanced safety in nursing homes by achieving **96%** accuracy in detecting falls and abnormal movements through the development of a robotic monitoring system using **Python**, **OpenCV**, and **ROS2**.
- Boosted system performance by **56%** and enabled seamless Intel RealSense camera integration by overhauling a legacy ROS2 codebase to meet the latest framework standards.
- Validated software functionality and ensured high accuracy across diverse nursing home environments by conducting extensive field testing, including real time motion tracking to simulate real world conditions and edge cases.
- Promoted efficient team progress across the full software development cycle by leading weekly collaboration meetings, aligning technical goals, addressing blockers, and ensuring consistent communication among team members.

Simulated Embedded Device Kernel | *C++*

[GitHub](#)

- Simulated a cooperative real-time operating system, enabling task scheduling, inter-process communication, and hardware abstraction, by designing a modular kernel architecture using **C++**.
- Developed custom hardware abstraction layers for GPIO, UART, and ADC devices, simulating embedded I/O behavior and enabling modular peripheral interaction using **object-oriented** design.

SKILLS

Languages: C++, Python, Golang, Rust, Assembly, Typescript, Verilog

Libraries/Frameworks: MFC, React, Node.js, TensorFlow, CUDA, RestAPI, MySQL, AWS

Developer Tools: Git, Docker, Linux, Visual Studio, Jenkins, FPGA